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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

9648/01

Paper 1 Multiple Choice Questions

29 September 2015

1 hour 15 mins

Additional Material: OTAS sheet

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A MCQ [40 marks]

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

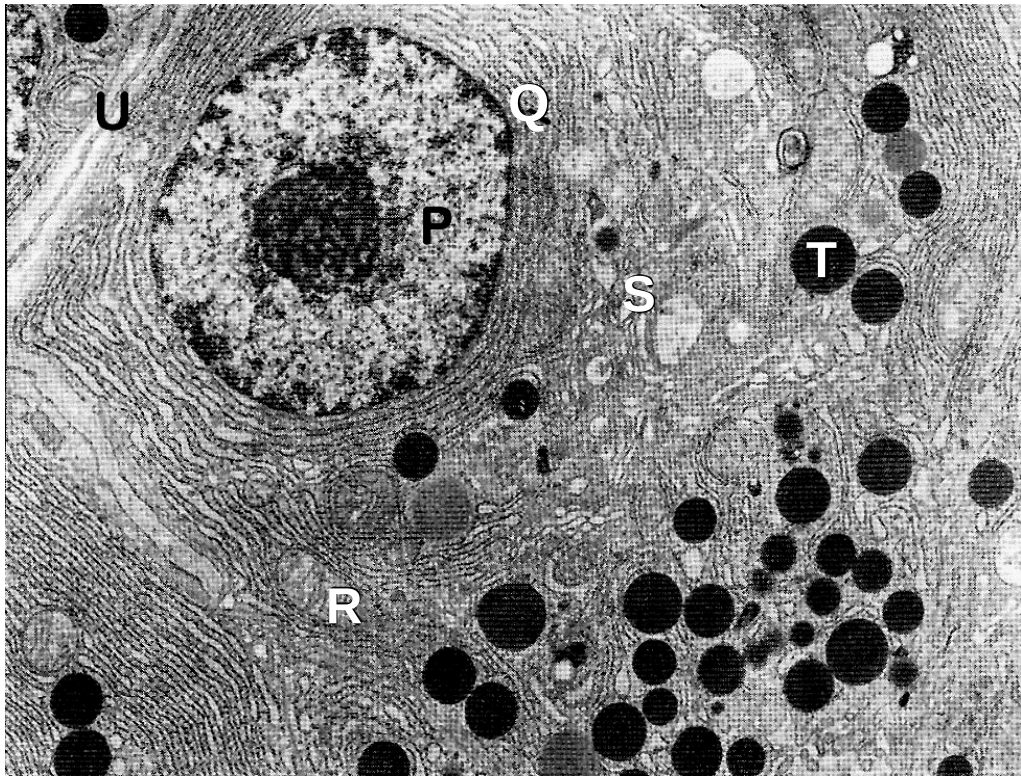
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This document consists of **28** printed pages and **0** blank page.

Answer **all** questions in this section.

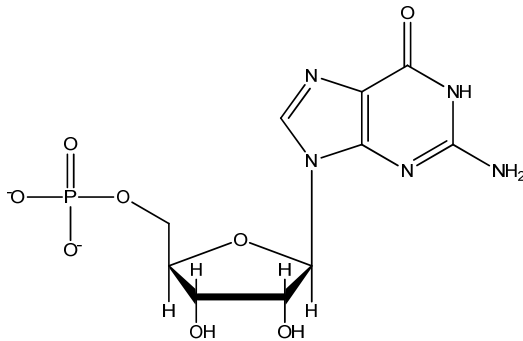
- 1 The figure below is an electron micrograph of a human cell.



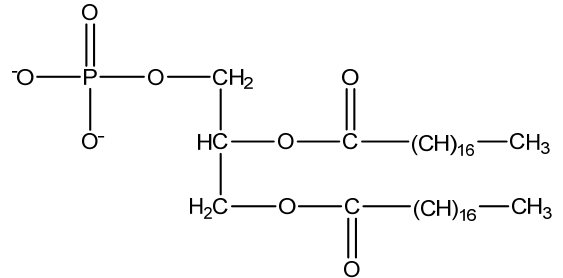
Which of the following correctly identifies its structure-function adaptation?

	structural adaptation	function
A	prominent P	antibody production
B	extensive Q, S and T	enzyme secretion
C	extensive R	phagocytosis
D	extensively-folded U	absorption

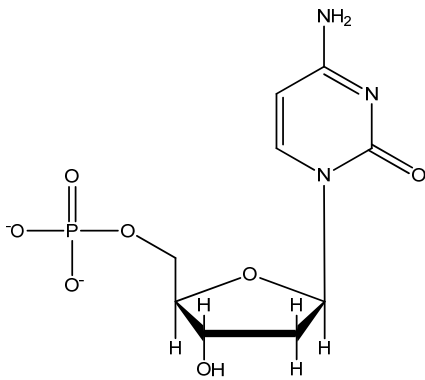
- 2 A student uses centrifugation to separate the various subcellular structures of human epithelial cells by size and density. Which of the following molecule(s) would you expect to find in the pellet containing the cell membrane?



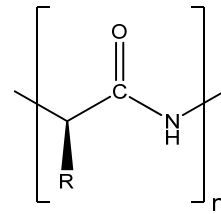
I



II



III



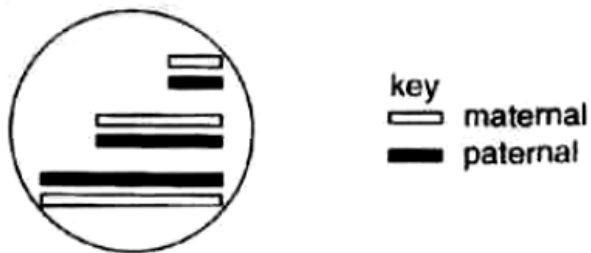
IV

- A II only
 B III only
 C II and IV only
 D I, III and IV only

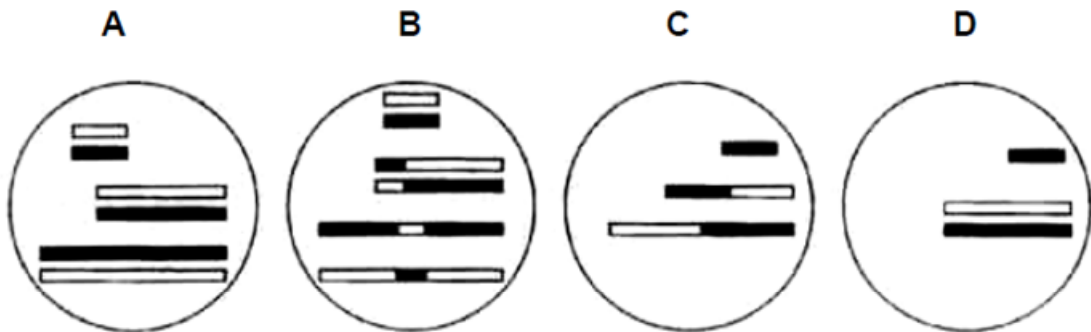
- 3 An enzyme is a globular protein held together by several different types of chemical bond, giving the enzyme primary, secondary and tertiary levels of structure. What of the following correctly summarises the types of bond involved in each level of structure?

	disulfide bonds	hydrogen bonds	ionic bonds	peptide bonds
A	tertiary	secondary, tertiary	tertiary	primary
B	primary, tertiary	primary	secondary, tertiary	secondary
C	secondary	secondary	tertiary	primary
D	primary	tertiary	secondary	tertiary

- 4 The diagram shows the maternal and paternal chromosomes from a diploid cell.

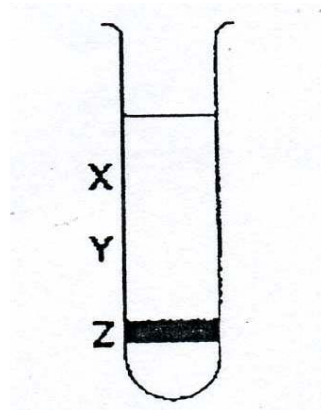


If the cell divides by meiosis, which diagram shows a possible viable gamete?



- 5 A culture of bacteria had all its DNA labelled with the heavy isotope of nitrogen, ^{15}N . The culture was then allowed to reproduce using nucleotides containing normal ^{14}N . The DNA was examined using a centrifuge after one generation and again after two generations.

The diagram shows the position of the DNA band at Z in the centrifuge tube when the DNA was first labelled.



In which pattern would the DNA be found after the first and after the second cell generations?

	After first generation	After second generation
A	Half at X and half at Y	Quarter at X and at Z and half at Y
B	Half at X and half at Z	Quarter at X and at Z and half at Y
C	All at X	Half at X and half at Y
D	All at Y	Half at X and half at Y

6 Which one of the following is the complementary sequence of 5'-TTAAGCGTAC-3'?

- A 5'-GTACGCTTAA-3'
- B 5'-AATTCGCATG-3'
- C 5'-CATGCGAATT-3'
- D 5'-TTAAGCGTAC-3'

7 Part of the amino acid sequences in normal and sickle cell haemoglobin are shown.

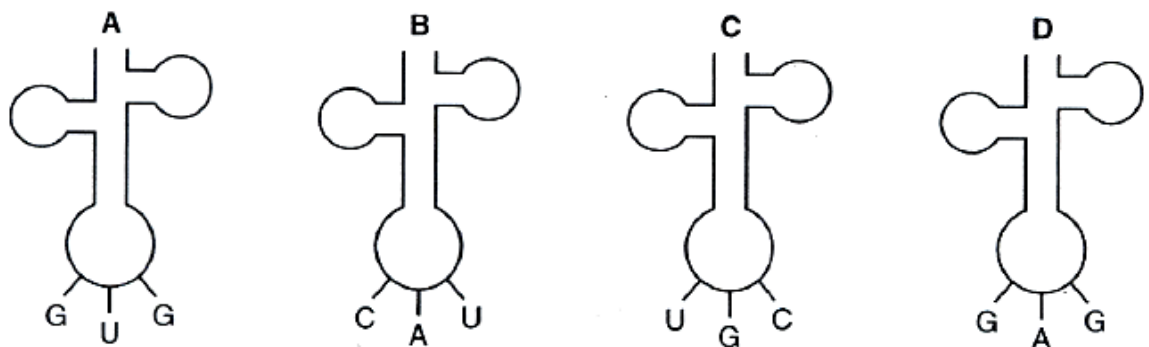
normal haemoglobin
thr-pro-glu-glu

sickle cell haemoglobin
thr-pro-val-glu

mRNA codons for these amino acids are:

Name of amino acid	mRNA codons	
Glutamine (glu)	GAA	GAG
Threonine (thr)	ACU	ACC
Proline (pro)	CCU	CCC
Valine (val)	GUA	GUG

Which transfer RNA molecule is involved in the formation of this part of the sickle cell haemoglobin?

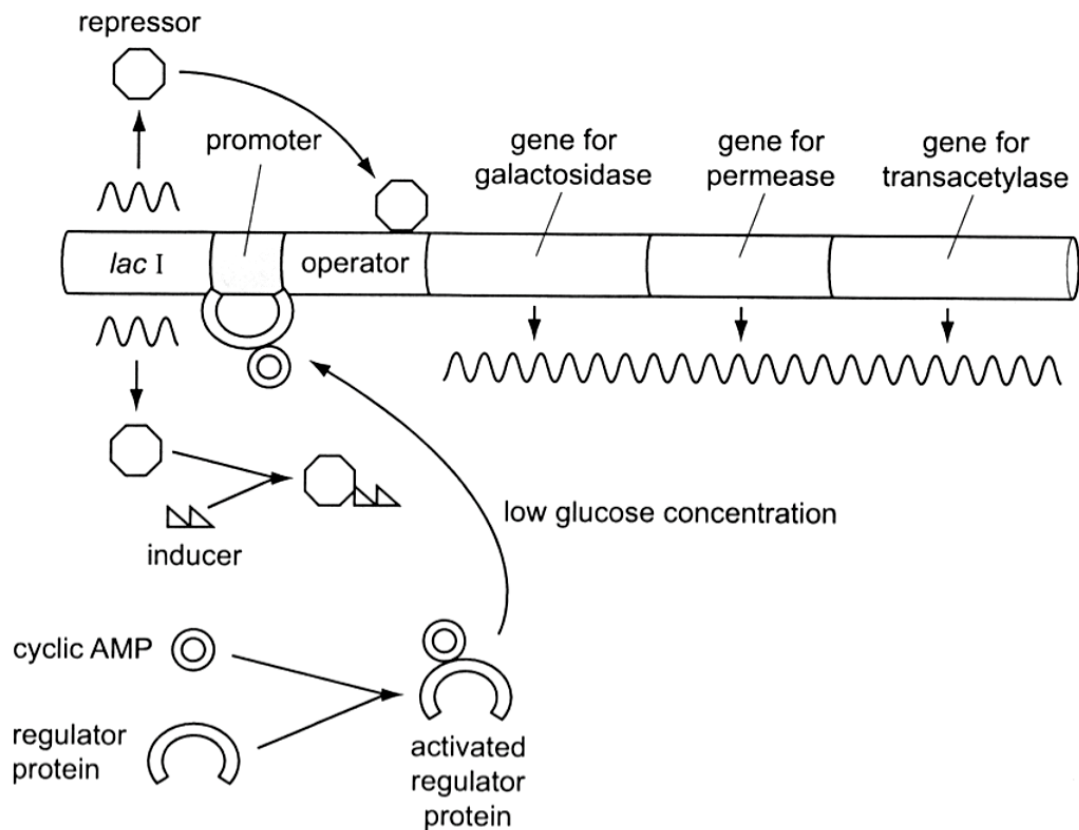


- 8 It was discovered in an alien species that their DNA contained two types of nitrogenous bases and there were 30 amino acids. What would be the minimum number of bases per codon that could code for the amino acids?
- A 3
 - B 5
 - C 6
 - D 8
- 9 Which feature occurs in the life cycle of the influenza virus?
- A Host cell DNA is destroyed by lytic enzymes.
 - B Viral genome is integrated into the host genome.
 - C Viral DNA acts as a template for DNA synthesis.
 - D Viruses enter the host cell by receptor-mediated endocytosis.
- 10 Some events that take place during generalised transduction are listed below.
- 1 Bacterial host DNA is fragmented
 - 2 Bacterial DNA may be packaged in a phage capsid
 - 3 Recombination between donor bacterial DNA and recipient bacterial DNA
 - 4 Phage infects a bacterial cell
 - 5 Phage DNA and proteins are made

Which sequence of events is **correct**?

- A 4 1 3 5 2
- B 4 1 5 2 3
- C 4 3 1 5 2
- D 4 5 1 3 2

11 The diagram shows some of the factors affecting the lac operon.



- 12** Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein product of that gene?

	Promoter	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Protein product is normal	Protein product is shorter than normal	No protein product is produced

- 13** Which combination of factors will affect translational control of gene expression?

	presence of 5' cap on mRNA	presence of carbohydrate group attached to polypeptide	presence of poly(A) tail at 3' end of mRNA	presence of repressor bound to 5' UTR of mRNA	presence of ubiquitin attached to polypeptide
A	no	yes	no	no	yes
B	no	yes	no	yes	yes
C	yes	no	yes	yes	no
D	yes	no	yes	no	no

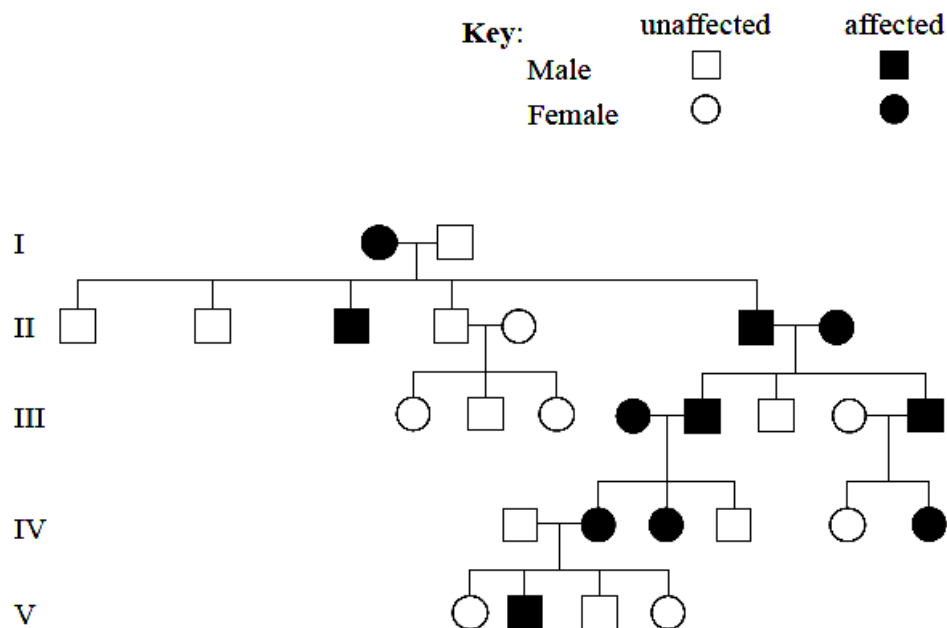
14 The following events occur in the extension of telomeres.

1. Further extension of 3' end of telomere.
2. Translocation of telomerase to end of telomere.
3. Reverse transcription to extend the 3' end of the telomere.
4. Complementary base-pairing of RNA template with single-stranded end of telomere.
5. DNA polymerase catalyses formation of complementary strand to form double-stranded DNA.

In which order do these events take place?

- A 3 2 4 5 1
 B 3 4 2 1 5
 C 4 2 3 5 1
 D 4 3 2 1 5

15 The pedigree chart below shows the inheritance of a genetic disease in a family. What is the nature of the allele that causes this disease?



- A Dominant and sex linked
 B Dominant and non-sex linked
 C Recessive and sex linked
 D Recessive and non-sex linked

- 16** An insect collection contains 102 specimens of a species of butterfly. This species is sexually dimorphic, meaning that the males and females look different from each other. A student examined the specimens and collected the following data.

Observation	Frequency
Blue wing colour (male)	64
Brown wing colour (female)	38
Wing span 35-37 mm	15
Wing span 37-39 mm	68
Wing span 39-41 mm	19

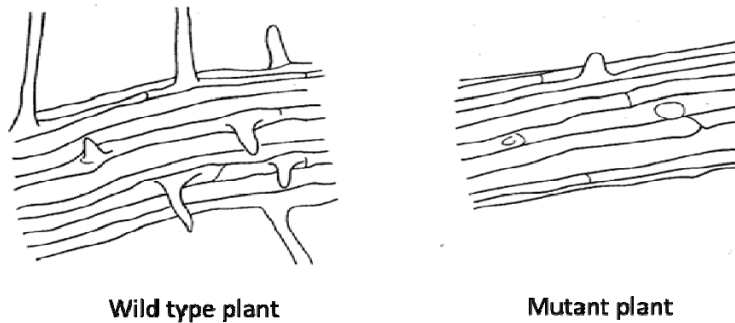
How should these observations be classified?

	Continuous variation	Discontinuous variation
A	Sexual dimorphism	Colour
B	Colour	Sexual dimorphism
C	Sexual dimorphism	Wing span
D	Wing span	Sexual dimorphism

- 17** Two plants of genotype **DdEeff** were crossed. Assuming independent assortment and simple dominance, what proportion of the offspring is expected to show the same phenotypic appearance as the parents?

- A** 1 out of 4
- B** 3 out of 4
- C** 3 out of 8
- D** 9 out of 16

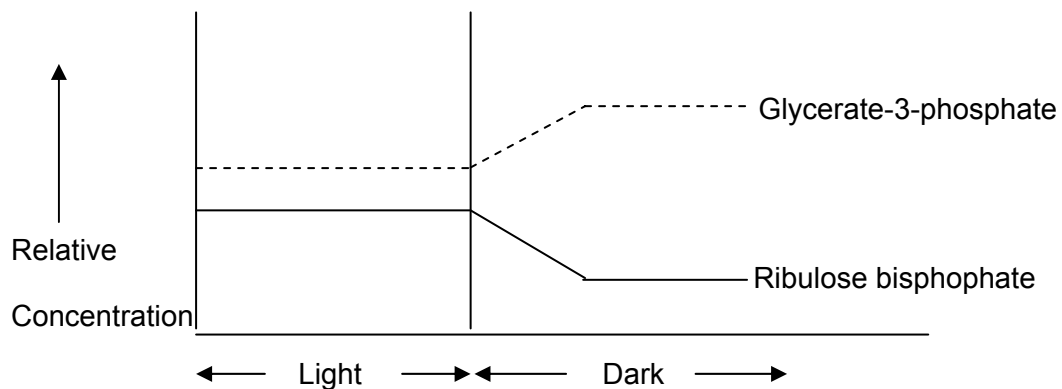
- 18** A mutant form of the plant *Arabidopsis thaliana* was discovered to have very short root hairs. The mutant contains a mutation in the gene coding for the enzyme reduced NADP oxidase. The normal enzyme converts reduced NADP (NADPH) to NADP^+ .



Which of the following could account for the presence of very short root hairs in the mutant?

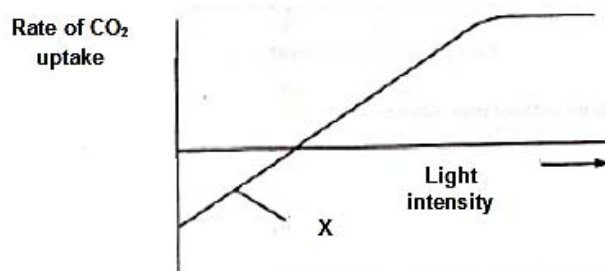
- A** As NADPH cannot be oxidised, rate of light-dependent reactions will decrease and cellular growth will also decrease.
- B** As NADPH cannot be oxidised, rate of light-dependent reactions will decrease and less water is needed for photolysis, hence root hairs can be shorter.
- C** As NADP^+ cannot be reduced, rate of light-independent reactions will decrease and cellular growth will also decrease.
- D** As NADP^+ cannot be reduced, rate of light-independent reactions will decrease and less water is needed for photolysis, hence root hairs can be shorter.

- 19 The graph below shows the effects of a change from light to dark conditions on the relative concentrations of glycerate-3-phosphate (GP) and ribulose biphosphate (RuBP) in a plant.



Which of the following features of the graph does **not** support the hypothesis that RuBP is converted to GP? The compounds _____.

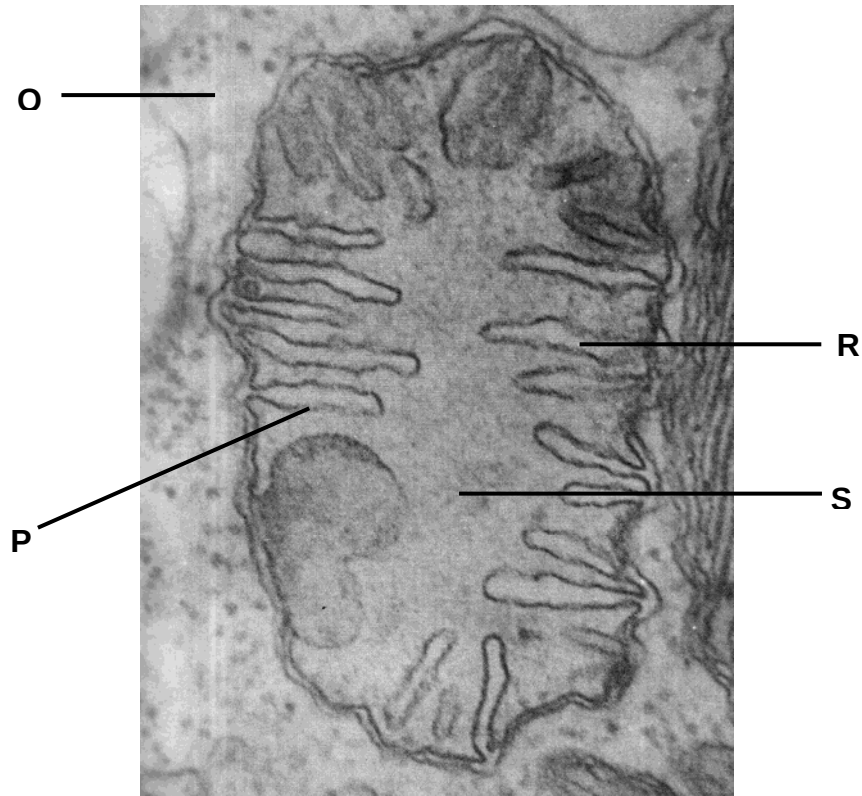
- A respond at the same time to the changed condition
 - B have constant concentrations at the start of the experiment
 - C achieve a new steady concentration at the same time
 - D have the same rate of response, one negative and the other positive
- 20 In the graph below, the rate of CO₂ uptake by plant cells is shown to vary with increasing light intensity.



Which of the following is true at point X?

- A The plant is photosynthesising.
- B Rate of respiration equals rate of photosynthesis.
- C CO₂ is a limiting factor.
- D There is not enough light for photosynthesis to have commenced.

21 The figure below shows an electron micrograph of an organelle.



Match the following processes with the structures labelled P – S above.

	Breakdown of fructose-6-phosphate	Oxidative phosphorylation only	Temporary lowering of pH	Formation of reduced co-enzymes
A	Q	R	S	P
B	R	S	P	Q
C	S	P	Q	R
D	Q	P	R	S

22 What would be the effect of inhibition of lactate dehydrogenase in a mammalian cell under anaerobic conditions?

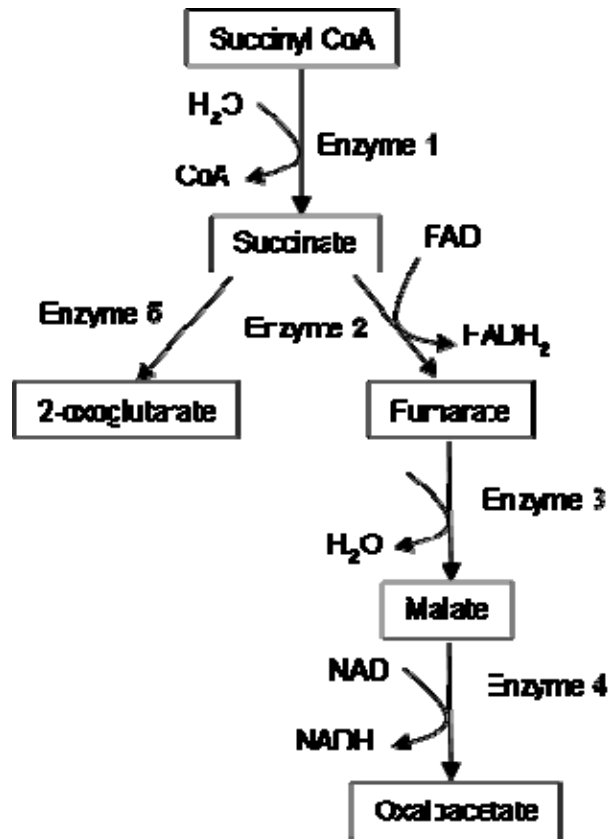
- A An increase in ATP production, due to increased amounts of reduced NAD.
- B A decrease in cell pH, due to the accumulation of lactic acid.
- C A decrease in glycolysis, due to the lack of NAD^+ .
- D An increase in the activity of Krebs cycle, due to increased amounts of pyruvate.

23 Which one of the following substances, when added, would directly result in a decline in ATP production in glycolysis?

- I** A chemical that would bind to NAD^+ irreversibly and induces its reduction to NADH.
- II** An inhibitor that has a similar structure to glucose but cannot be broken down by respiratory enzymes.
- III** A chemical that creates an anaerobic environment by combusting in oxygen
- IV** A reagent that binds to the active site of ATPase permanently

- A** I and II only
- B** III and IV only
- C** I, II and IV only
- D** All of the above

- 24 A competitive inhibitor of the enzyme catalyzing the oxidative decarboxylation of α -ketoglutarate (5C) was added to respiring animal tissues.



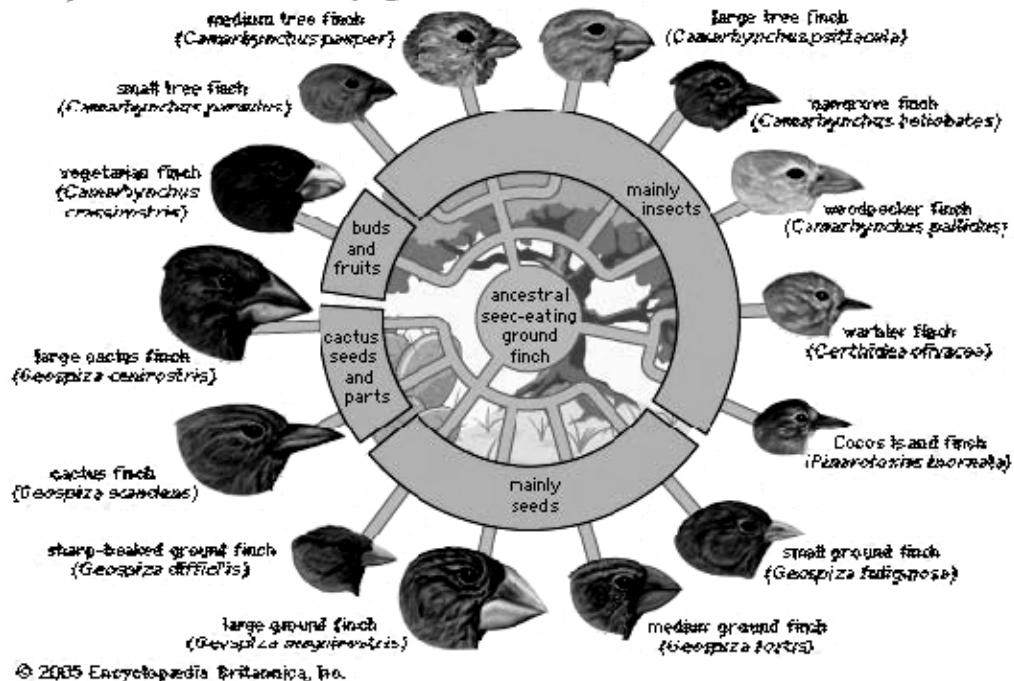
Which will occur in the presence of a high concentration of oxaloacetate?

- 1 Positive feedback will occur, resulting in the production of more fumarate and malate.
- 2 Oxidative phosphorylation will cease.
- 3 Krebs cycle will proceed due to a high concentration of oxaloacetate to form less 2-oxoglutarate.
- 4 Higher rates of link reaction will occur with oxidative decarboxylation taking place in the mitochondrial matrix to produce more acetyl-CoA.

- A 2 only
 B 2 and 3
 C 1 and 4
 D 1, 3 and 4

- 25 The Galapagos Islands are a group of volcanic islands in the eastern Pacific Ocean, about 1000km from South America. The diagram below illustrates the finches found on the islands; they resemble each other closely but differ in their feeding habits and in the shape of their beaks.

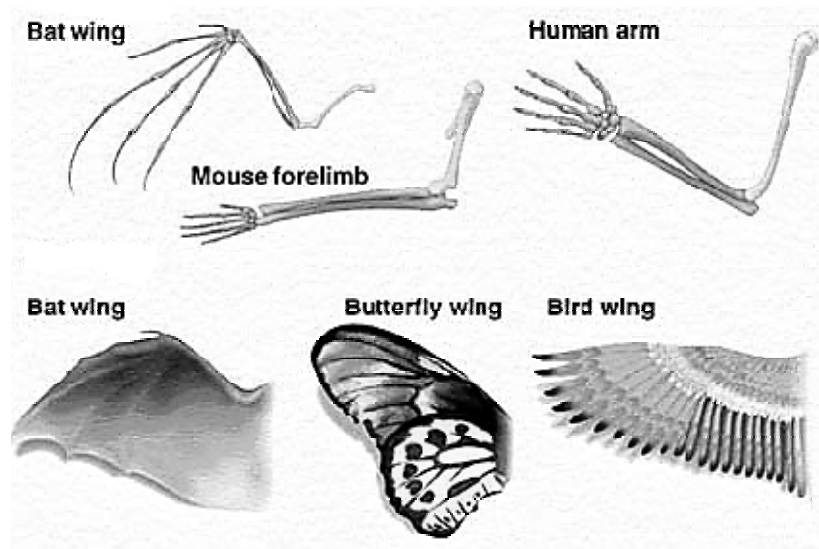
Adaptive radiation in Galapagos finches



Which statement below explains how so many distinct and yet similar species arose?

- A As a result of the different food sources available, the ancestral species acquired different mutations which resulted in the emergence of a variety of beak shapes.
- B The finches were subjected to similar selection pressures which resulted in similar phenotypic characteristics in the distinct populations.
- C The difference in food sources served as a selection pressure, allowing only finches with certain beak shapes to survive and reproduce.
- D The finches migrated only to environments that they are already adapted to; finches with different phenotypes migrated to different islands

- 26 The diagram below shows the bone structures of the human arm, mouse forelimb and bat wings as well as the morphology of the wings of bat, butterfly and birds.

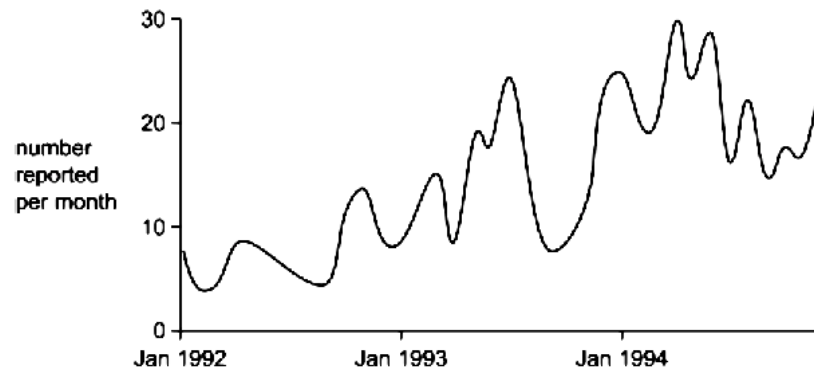


Which of the following are correct conclusions made from the diagram provided?

- 1 Bat, mouse and human share a common ancestor as their bone structure exhibit anatomical homology.
- 2 Variations in the bone morphology of bats, mouse and human are due to natural selection.
- 3 Bat, butterfly and bird share a recent common ancestor as shown by their common wing morphology.
- 4 Bat, butterfly and bird exhibit analogous structures.

- A 1 and 2
 B 2 and 3
 C 1 and 4
 D 1, 2 and 4

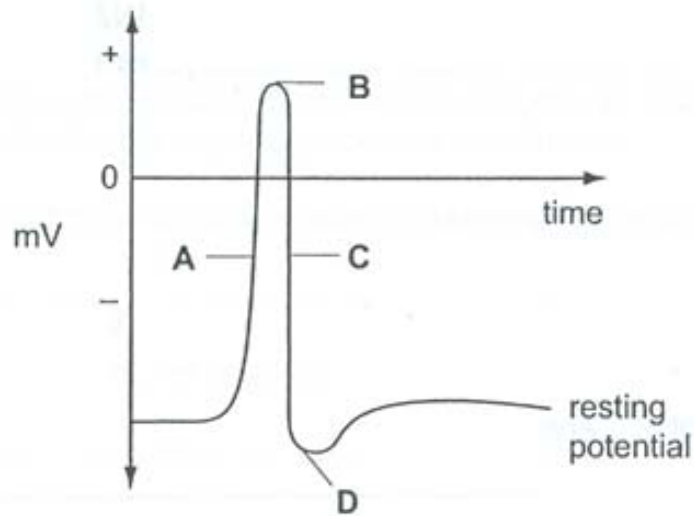
- 27 The graph shows the change in the number of bacterial samples from some New York hospitals were resistant to the antibiotic vancomycin in 1992 to 1994. Forty samples were taken each month from randomly selected patients who had become infected with bacteria in the hospital.



Which statement most accurately describes the cause of the changes in the frequency of the vancomycin resistant phenotype?

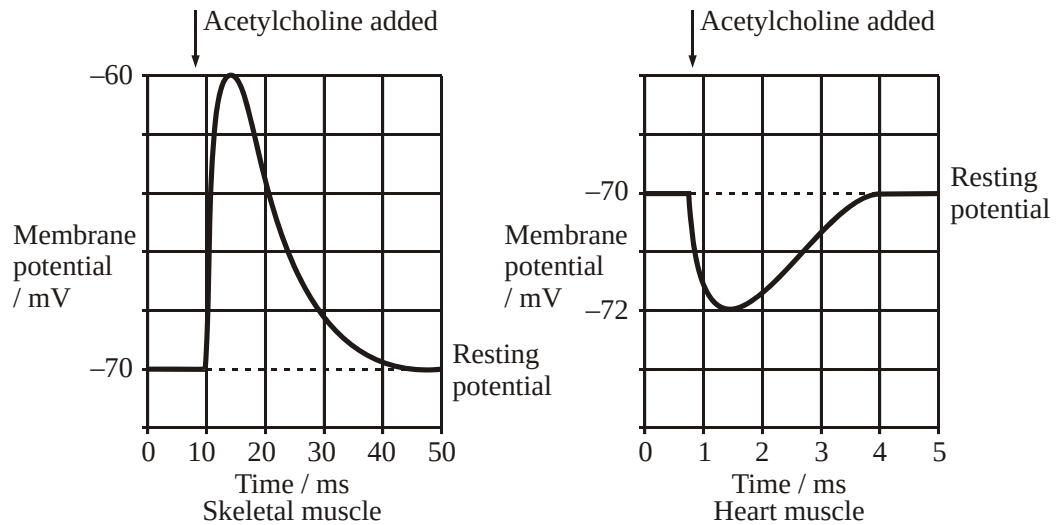
- A effect of artificial selection
- B effect of natural selection
- C purely due to genetic drift
- D purely due to founder's effect

28 The diagram shows an action potential. Which of the following statements is **correct**?



- A At point A, the axon is impermeable to sodium ions.
- B At point B, the peak of the action potential can be reduced by decreasing the external K^+ concentration.
- C At point C, the voltage-gated sodium channels are closed and ligand-gated potassium channels are open.
- D At point D, the action potential travels in one direction due to the refractory period.

- 29 The graphs below show the effect of adding acetylcholine to skeletal muscle and heart muscle.



Which of the following statements are true of the results above?

- 1 Depolarisation occurred when acetylcholine was added to skeletal muscle but hyperpolarisation occurred when acetylcholine was added to heart muscle.
- 2 Sodium channels opened when acetylcholine was added to skeletal muscle but potassium channels opened when acetylcholine was added to heart muscle.
- 3 Acetylcholine makes it easier to result in contraction of the skeletal muscle but harder to result in heart muscle contraction.
- 4 Binding of acetylcholine leads to formation of IPSP in skeletal muscle and EPSP in heart muscle.

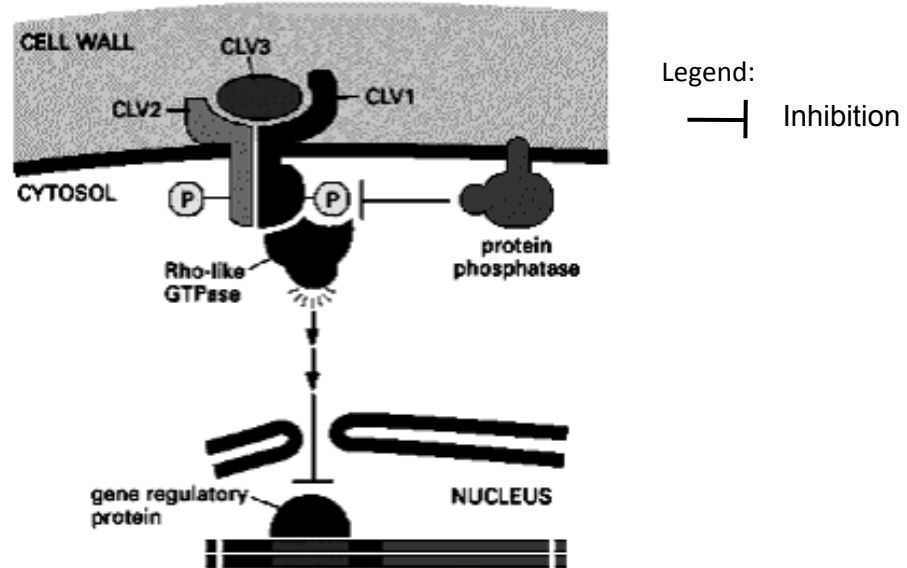
- A 1 and 2
 B 2 and 4
 C 1, 2 and 3
 D All of the above

30 Saltatory conduction is faster than non-saltatory conduction because

- 1** myelinated neurons have a lower internal resistance than do non-myelinated ones, thereby leading to faster conduction.
- 2** the insulation provided by myelin sheath forces the depolarizing current farther down the axon, thereby allowing the action potential to skip parts of the membrane.
- 3** myelinated axons are bigger than non-myelinated ones, and larger diameter axons conduct faster.
- 4** there are higher concentration of sodium channels at the nodes which generates much larger action potential than the non-myelinated axon, thus, generate faster conduction.

- A** 2 only
- B** 4 only
- C** 1 and 2
- D** 3 and 4

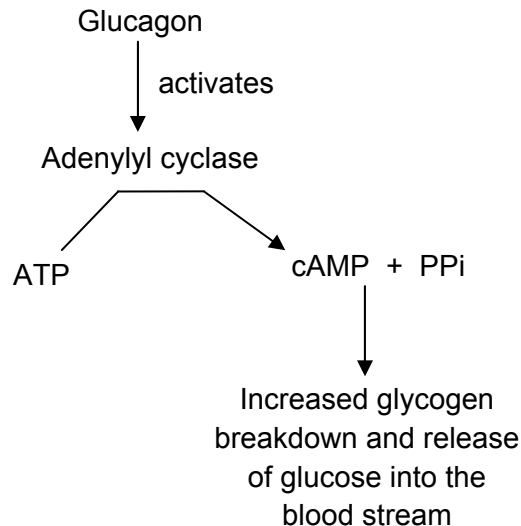
- 31 The diagram below shows the cell signalling pathway involved in cell proliferation and differentiation in the shoot meristem.



Which one of the following sequences regarding the activation of the above signalling pathway is **correct**?

- A binding of CLV3 to subunits CLV2 and CLV1 → dimerisation of CLV2 and CLV1 → autophosphorylation of serine/threonine residues → activation of Rho-like protein → inhibition of gene transcription
- B dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription
- C binding of CLV3 to subunits CLV2 and CLV1 → dimerisation of CLV2 and CLV1 → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → inhibition of gene transcription.
- D dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → inhibition of gene transcription

- 32** One action of the hormone glucagon is to cause the activation of adenylyl cyclase in liver cells. Adenylyl cyclase is an enzyme that converts ATP into the messenger molecule cyclic AMP (cAMP), the presence of which triggers a cascade of events within the cell that leads to the breakdown of glycogen and the subsequent release of glucose into the blood stream. This pathway is summarised below.



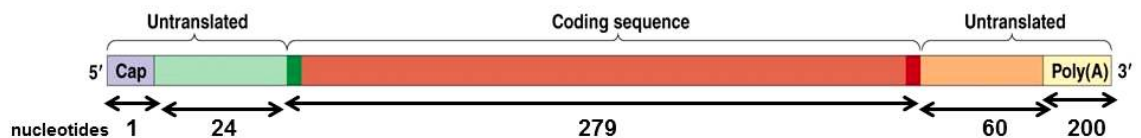
When glucagon is removed, adenylyl cyclase becomes inactive again and ceases to convert ATP to cAMP. The enzyme phosphodiesterase converts cAMP into AMP. The following table summarises an experiment measuring glucose release by liver cell cultures under various conditions.

Culture	Condition(s)	Observation(s)
1	no addition of glucagon and caffeine	minimal glucose release
2	glucagon added and later removed	glucose released into media; rapid reduction in release after glucagon removed
3	caffeine added	minimal glucose release
4	glucagon and caffeine added; glucagon was then removed but caffeine maintained	glucose released into media; slow reduction in release after glucagon removed

Based on these observations, which of the following statement is most likely to be true?

- A Caffeine stimulates synthesis of cAMP.
- B Caffeine inhibits the action of phosphodiesterase.
- C Caffeine binds the active site of adenylyl cyclase.
- D Caffeine prevents glucagon breakdown by insulin.

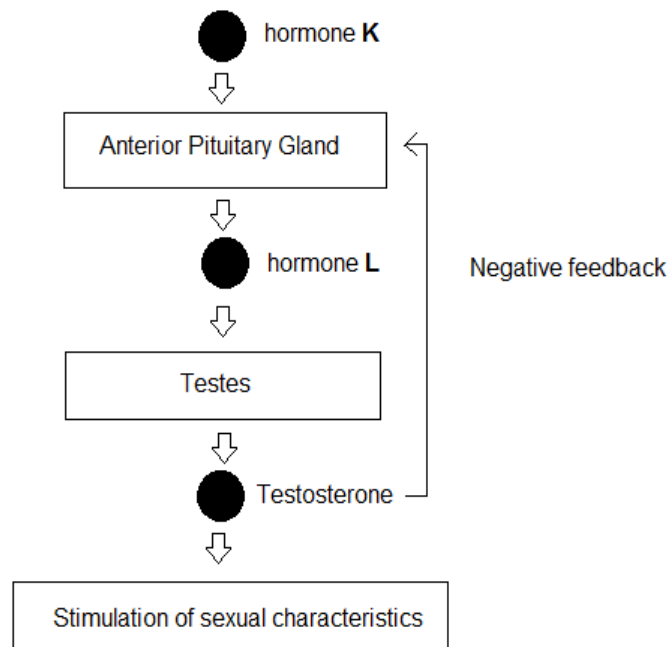
- 33** The restriction enzyme *DdeI* recognizes a sequence on DNA, CTNAG (where N is any base). A portion of the β -haemoglobin sequence encoding normal haemoglobin is CCTGAGGAG. If the mutation leading to sickle cell anaemia had been CCTCAGGAG, would you be able to distinguish the two alleles with a restriction digest?
- A** Yes, the normal allele will be digested, the sickle allele will not.
- B** Yes, the sickle allele will be digested, the normal allele will not
- C** No, both will be digested by *DdeI*.
- D** No, neither will be digested by *DdeI*.
- 34** Which of the following is not a goal of the Human Genome Project?
- A** Identify all alleles present in human beings.
- B** Map the genetic differences between individuals.
- C** Map and sequence all non-coding sequences in human beings.
- D** Develop computer software to analyse genetic information which had been sequenced.
- 35** The mature eukaryotic mRNA contains the structure shown below.



When the reverse transcription of this mRNA is performed for insertion into a bacterial plasmid for transforming *E. coli* cells, the length of the resulting cDNA fragment will be _____.

- A** 279 bp
- B** 303 bp
- C** 363 bp
- D** 564 bp

- 36 The production of testosterone is regulated by the anterior pituitary gland. Upon stimulation by hormone K, the gland releases hormone L which causes the testes to produce testosterone. The elevated level of testosterone exerts a negative feedback on the anterior pituitary gland.



If an individual consumes large amount of steroids containing testosterone, which of the following would be observed?

	Production of hormone K	Production of hormone L	Production of testosterone	Stimulation of sexual characteristics
A	Decrease	Decrease	Decrease	No change
B	Decrease	Decrease	No change	Increase
C	No change	Decrease	Decrease	Increase
D	No change	Decrease	Decrease	No change

37 Which of the following statements is/are true regarding zygotic stem cells and cancer cells?

- I** Both are able to move one from location to another.
- II** Both are able to divide by mitotic division.
- III** Both are specialised cells and capable to differentiate further.
- IV** Both are capable of indefinite replication.

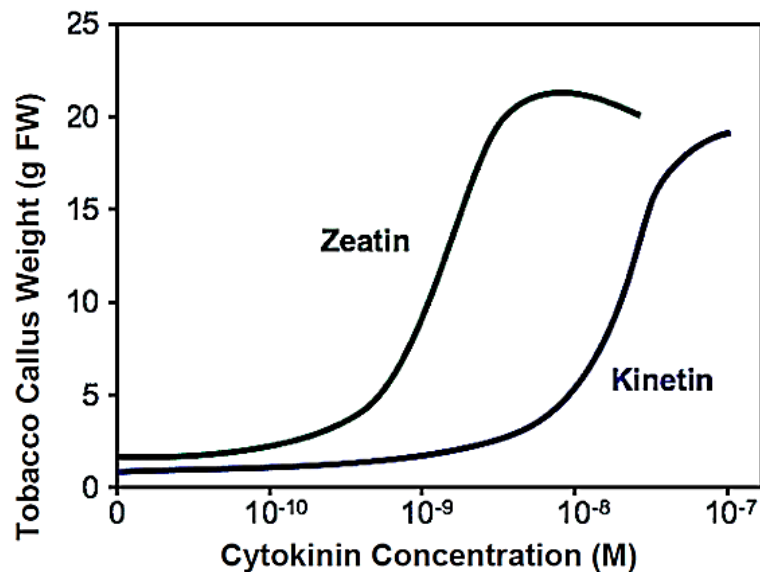
- A** II only
- B** I and III
- C** II and IV
- D** I, II and IV

38 One therapy to treat β -thalassaemia is to transplant bone marrow cells from a genetically compatible donor into a patient. A potential gene therapy involves adding the normal, dominant allele for β -globin to the patient's cells.

Which of the following would ensure that the normal gene is passed on to the next generation?

- A** Using an adenoviral vector to introduce the normal β -globin gene into bone marrow cells.
- B** Using an adenoviral vector to introduce the normal β -globin gene into an egg cell.
- C** Using a retrovirus to introduce the normal β -globin gene into bone marrow cells.
- D** Using a retrovirus to introduce the normal β -globin gene into an egg cell.

- 39 The graph below shows the effect of natural cytokinin (zeatin) and synthetic cytokinin (kinetin) on tobacco callus growth.



Which of the following statements can be concluded from the graph?

- A Intermediate concentrations of both kinetin and auxin were required for callus growth.
 - B Natural cytokinin is more effective at lower concentrations than artificial cytokinin in inducing tobacco callus growth.
 - C For callus growth, the optimum concentration for kinetin is 10^{-7} M and the optimum concentration for zeatin is 10^{-8} M.
 - D The tobacco callus growth increases with increasing zeatin concentration.
- 40 Which of the following is **not** an example of genetically modified organisms?
- A AquaAdvantage Salmon that grow to adult size quickly.
 - B Bt corn plants that are insect-resistant.
 - C Golden Rice that produce high levels of beta-carotene.
 - D Milk-producing cows injected with bovine somatotrophin (BST).

END OF PAPER

2015 Y6 Preliminary Exam H2**MCQ Answer Scheme**

1	B	21	D
2	C	22	C
3	A	23	A
4	C	24	A
5	D	25	C
6	A	26	D
7	B	27	B
8	B	28	D
9	D	29	C
10	B	30	A
11	C	31	A
12	B	32	B
13	C	33	C
14	D	34	B
15	B	35	D
16	D	36	C
17	D	37	C
18	A	38	D
19	B	39	B
20	A	40	D